

REGRESSION AND CORRELATION

REGRESSION:

Dependence of one variable (*called dependent variable*) on the other variable or variables (*called independent variables*) is called as Regression.

e.g. Consumption depends upon income, yield of wheat depends upon labor, seeds, fertilizers, fertility of soil etc.

Types of Regression:

There are two types of Regression.

- (i) Simple Regression
- (ii) Multiple Regression

Simple Regression:

Dependence of one variable on the other one variable is called as Simple Regression.
e.g. Consumption depends upon income etc.

Multiple Regression:

Dependence of one variable on the other variables is called as Multiple Regression.
e.g. Yield of wheat depends upon labor, seeds, fertilizers, fertility of soil etc.

Dependent variable and Independent variable:

The variable which we want to estimate or predict is called as **Dependent variable**

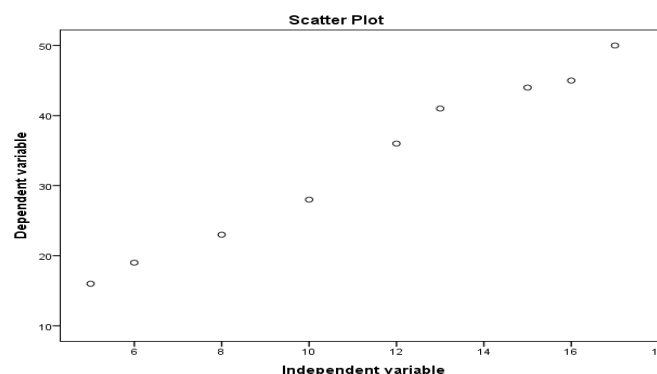
The variable on the basis of which the dependent variable is to be estimated is called as **Independent variable**.

e.g. we want to estimate consumption on the basis of income, then Consumption is **dependent variable** and Income is **independent variable**.

Dependent variable is also called as Regressand, the Predictand, the Explained, the Response variable or random variable and the independent variable is also called as Regressor, The predictor, the Explanatory or the Fixed variable.

Scatter diagram:

If we plot the independent variable along X-axis and the corresponding depend variable along Y-axis, then the resulting set of points is called Scatter diagram.



Simple Linear Regression:

A regression line represented by straight line is called Simple Linear Regression.

The estimated Simple Linear Regression equation of Y on X OR The estimated equation of least square line of Y on X OR The equation of Straight Line of Y on X:

$$\hat{Y} = a + bX$$

Where Y = dependent variable; X = Independent variable

a = Intercept;

b = slope

$$b = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sum (X - \bar{X})^2}$$

OR
$$b = \frac{\sum XY - \frac{\sum X \sum Y}{n}}{\sum X^2 - \frac{(\sum X)^2}{n}}$$

OR
$$b = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$

OR
$$b = \frac{\sum XY - n \bar{X} \bar{Y}}{\sum X^2 - n (\bar{X})^2}$$

And
$$a = \bar{Y} - b \bar{X}$$

Note:

$b = b_{YX}$ = Regression coefficient of Y on X

Note:

b measures average change in Y for a unit change in X.

Error Or Residual:

$$e_i = Y_i - \hat{Y}_i$$

Where Y = actual values;

$\hat{Y}$ = estimated values.

Note: (i) $\sum e_i = \sum (Y_i - \hat{Y}_i) = 0$

(ii) $\sum Y = \sum \hat{Y}$

(iii) $\bar{Y} = \bar{\hat{Y}}$

Example# 1: Compute the least squares regression equation of Y on X for the following data. What is the regression coefficient and what does it mean?

X	5	6	8	10	12	13	15	16	17
Y	16	19	23	28	36	41	44	45	50

Solution:

X	Y	XY	X^2
5	16	80	25
6	19	114	36
8	23	184	64
10	28	280	100
12	36	432	144
13	41	533	169
15	44	660	225
16	45	720	256
17	50	850	289
$\Sigma X = 102$	$\Sigma Y = 302$	$\Sigma XY = 3853$	$\Sigma X^2 = 1308$

The estimated least squares regression equation of Y on X is

$$\hat{Y} = a + bX$$

$$\bar{X} = \frac{\Sigma X}{n} = \frac{102}{9} = 11.33; \quad \bar{Y} = \frac{\Sigma Y}{n} = \frac{302}{9} = 33.56$$

$$b = \frac{n \Sigma XY - \Sigma X \Sigma Y}{n \Sigma X^2 - (\Sigma X)^2}$$

$$b = \frac{9(3853) - (102)(302)}{9(1308) - (102)^2} = \frac{34677 - 30804}{11772 - 10404} = \frac{3873}{1368}$$

$$b = 2.831$$

$$a = \bar{Y} - b\bar{X}$$

$$a = 33.56 - (2.831)(11.33)$$

$$a = 1.47$$

Therefore fitted equation is

$$\hat{Y} = 1.47 + 2.831X$$

Regression Coefficient = $b = 2.831$

Interpretation

$b = 2.831$ indicates that Y averagely increase 2.831 for a unit change in X .

Example# 2: Given these ten pairs of (X, Y) values

X	1	1	2	3	4	4	5	6	6	7
Y	2.1	2.5	3.1	3.0	3.8	3.2	4.3	3.9	4.4	4.8

- Find the estimated simple linear regression equation $\hat{Y} = a + bX$.
- Compute the residuals and verify that they add to zero.
- Use the regression equation to predict the values of Y when $X = 10$

P.T.O.

Solution:

				(b)	
X	Y	XY	X^2	$\hat{Y} = 1.99 + 0.39X$	$e = Y - \hat{Y}$
1	2.1	2.1	1	$1.99 + 0.39(1) = 2.38$	- 0.28
1	2.5	2.5	1	$1.99 + 0.39(1) = 2.38$	0.12
2	3.1	6.2	4	$1.99 + 0.39(2) = 2.77$	0.33
3	3.0	9.0	9	$1.99 + 0.39(3) = 3.16$	- 0.16
4	3.8	15.2	16	3.55	0.25
4	3.2	12.8	16	3.55	- 0.35
5	4.3	21.5	25	3.94	0.36
6	3.9	23.4	36	4.33	- 0.43
6	4.4	26.4	36	4.33	0.07
7	4.8	33.6	49	4.72	0.08
$\Sigma X = 39$	$\Sigma Y = 35.1$	$\Sigma XY = 152.7$	$\Sigma X^2 = 193$		$\Sigma e = 0$

(a) The estimated least squares regression equation of Y on X is

$$\hat{Y} = a + bX$$

$$\bar{X} = \frac{\Sigma X}{n} = \frac{39}{10} = 3.9; \quad \bar{Y} = \frac{\Sigma Y}{n} = \frac{35.1}{10} = 3.51$$

$$b = \frac{n \Sigma XY - \Sigma X \Sigma Y}{n \Sigma X^2 - (\Sigma X)^2}$$

$$b = \frac{10(152.7) - (39)(35.1)}{10(193) - (39)^2}$$

$$b = 0.39$$

$$a = \bar{Y} - b\bar{X}$$

$$a = 3.51 - (0.39)(3.9)$$

$$a = 1.99$$

Therefore fitted equation is

$$\hat{Y} = 1.99 + 0.39X \dots\dots\dots(i)$$

(c) To predict the values of Y when $X = 10$, we put $X = 10$ in equation (i)

$$\hat{Y} = 1.99 + 0.39(10)$$

$$\hat{Y} = 5.89$$

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Example# 3: A study was made to know the relation between advertising expenditure (X) and the increase in sales (Y). Following data were obtained;

X	40	20	25	20	30	50	40	20	25
Y	85	100	95	175	140	190	120	120	180

- (i)** Find the regression line to predict increase in sales from advertising expenditure.

- (ii) Estimate increase in sales when advertising expenditure is Rs. 35/

Example# 4: The values of x and their corresponding values of y are shown in the table below;

x	0	1	2	3	4
y	2	3	5	4	6

- (i) Find the least square regression line $y = a + bx$
 (ii) Estimate the value of y when $x = 10$

Example# 5: Using the method of least squares, obtain the straight line and show that $\sum(y - \hat{y}) = 0$ for the given data.

x	12	14	16	18	20	22
y	6	9	11	8	10	12

Example# 6: The following are the final grades obtained by 8 students in Statistics and Psychology;

Statistics	74	93	55	41	23	92	64	40
Mass. Com.	81	86	67	35	30	100	55	52

Find the equation of line of regression which enables us to predict a student's final grade in Mass. Com. on the basis of his final grade in Statistics.

D.Y.S.

The estimated Simple Linear Regression equation of X on Y OR The estimated equation of least square line of X on Y OR The equation of Straight Line of X on Y :

$$\hat{X} = c + dY$$

Where X = dependent variable; Y = Independent variable
 c = Intercept; d = slope

$$d = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sum(Y - \bar{Y})^2}$$

$$\text{OR } d = \frac{\sum XY - \frac{\sum X \sum Y}{n}}{\sum Y^2 - \frac{(\sum Y)^2}{n}}$$

$$\text{OR } d = \frac{n \sum XY - \sum X \sum Y}{n \sum Y^2 - (\sum Y)^2}$$

$$\text{OR } d = \frac{\sum XY - n \bar{X} \bar{Y}}{\sum Y^2 - n(\bar{Y})^2}$$

And $c = \bar{X} - d\bar{Y}$

Note: $b = b_{YX}$ = Regression coefficient of Y on X

$d = b_{XY}$ = Regression coefficient of X on Y

Note: Both the regression coefficients always have the same signs for the same data.

Example# 7: Given the following sets of values

Y	6.5	5.3	8.6	1.2	4.2	2.9	1.1	3.0
X	3.2	2.7	4.5	1.0	2.0	1.7	0.6	1.9

(a) Compute the least squares regression equation for Y values on X values i.e.

$$\hat{Y} = a + bX$$

(b) Compute the least squares regression equation for X values on Y values i.e.

$$\hat{X} = c + dY$$

Solution:

Y	X	XY	X ²	Y ²
6.5	3.2	20.8	10.24	42.25
5.3	2.7	14.31	7.29	28.09
8.6	4.5	38.7	20.25	73.96
1.2	1.0	1.2	1.00	1.44
4.2	2.0	8.4	4.00	17.64
2.9	1.7	4.93	2.89	8.91
1.1	0.6	0.66	0.36	1.21
3.0	1.9	5.7	3.61	9.00
$\Sigma Y = 32.8$	$\Sigma X = 17.6$	$\Sigma XY = 94.7$	$\Sigma X^2 = 49.64$	$\Sigma Y^2 = 182$

(a) The estimated least squares regression equation of Y on X is

$$\hat{Y} = a + bX$$

$$\bar{X} = \frac{\Sigma X}{n} = \frac{17.6}{8} = 2.2;$$

$$\bar{Y} = \frac{\Sigma Y}{n} = \frac{32.8}{8} = 4.1$$

$$b = \frac{n \Sigma XY - \Sigma X \Sigma Y}{n \Sigma X^2 - (\Sigma X)^2} = \frac{8(94.7) - (17.6)(32.8)}{8(49.64) - (17.6)^2}$$

$$b = 2.06$$

$$a = \bar{Y} - b\bar{X}$$

$$a = 4.1 - (2.06)(2.2)$$

$$a = -0.43$$

Therefore fitted equation is

$$\hat{Y} = -0.43 + 2.06X$$

(b) The estimated least squares regression equation of X on Y is

$$\hat{X} = c + dY$$

$$d = \frac{n \Sigma XY - \Sigma X \Sigma Y}{n \Sigma Y^2 - (\Sigma Y)^2} = \frac{8(94.7) - (17.6)(32.8)}{8(182) - (32.8)^2}$$

$$d = 0.47$$

$$c = \bar{X} - d\bar{Y}$$

$$c = 2.2 - (0.47)(4.1)$$

$$c = 0.273$$

Therefore fitted equation is

$$\hat{X} = 0.273 + 0.47Y$$

COEFFICIENT OF DETERMINATION:

The ratio of the explained variation to the total variation is called as coefficient of Determination. It is denoted by r^2 for sample data.

$$r^2 = \frac{\text{Explained variation}}{\text{Total variation}}$$

$$r^2 = \frac{\sum(\hat{Y} - \bar{Y})^2}{\sum(Y - \bar{Y})^2}$$

$$r^2 = \frac{a \sum Y + b \sum XY - \frac{(\sum Y)^2}{n}}{\sum Y^2 + \frac{(\sum Y)^2}{n}}$$

Interpretation

e.g. $r^2 = 0.93$ indicates that there is 93% of the variability in Y (Dependent variable) is explained due to its linear relationship with X (independent variable) and remaining 7% is due to chance or error.

Note: $0 \leq r^2 \leq 1$

Example# 8: Compute the coefficient of determination for the following data and interpret the coefficient.

Income (X) (000)	10	20	30	40	50	60
Expenditure (Y) (000)	7	21	23	34	36	53

Solution:

X	Y	XY	X ²	Y ²
10	7	70	100	49
20	21	420	400	441
30	23	690	900	529
40	34	1360	1600	1156
50	36	1800	2500	1296
60	53	3180	3600	2809
$\sum X = 210$	$\sum Y = 174$	$\sum XY = 7520$	$\sum X^2 = 9100$	$\sum Y^2 = 6280$

$$\bar{X} = \frac{\sum X}{n} = \frac{210}{6} = 35; \quad \bar{Y} = \frac{\sum Y}{n} = \frac{174}{6} = 29$$

$$b = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$

$$b = \frac{6(7520) - (210)(174)}{6(9100) - (210)^2} = 0.82$$

$$a = \bar{Y} - b\bar{X}$$

$$a = 29 - (0.82)(35)$$

$$a = 0.30$$

Now

$$r^2 = \frac{a \sum Y + b \sum XY - \frac{(\sum Y)^2}{n}}{\sum Y^2 - \frac{(\sum Y)^2}{n}}$$

$$r^2 = \frac{(0.30)(174) + (0.82)(7520) - \frac{(174)^2}{6}}{6280 - \frac{(174)^2}{6}}$$

$$r^2 = \frac{52.2 + 6166.4 - 5046}{6280 - 5046}$$

$$r^2 = \frac{1172.6}{1234}$$

$$r^2 = 0.95$$

Interpretation

e.g. $r^2 = 0.95$ indicates that there is 95% of the variability in Y (Expenditures) is explained due to its linear relationship with X (income) and remaining 5% is due to chance or error or some other factors.

Example# 9: In an experiment to measure the stiffness of a spring, the length of the spring under different loads was measured as follows

X = Loads (lb)	3	5	6	9	10	12	15	20	22	28
Y = Length (in.)	10	12	15	18	20	22	27	30	32	34

Compute the coefficient of determination and interpret the coefficient. **D.Y.S.**

Properties of Least Square Regression Line:

(i) The least square regression line always passes through its mean points $(\bar{X}, \bar{Y})$

(ii) The sum of deviations of observations Y_i from $\hat{Y}$ is always equal to zero.

i.e. $\sum (Y - \hat{Y}) = 0$

(iii) The sum of squared deviations of observations Y_i from $\hat{Y}$ is minimum.

i.e. $\sum (Y - \hat{Y})^2 = \text{Minimum}$

(iv) The least square regression line is the line of best fit.

CORRELATION:

The interdependence between two variables is called correlation.

OR

Two variables are said to be correlated, if they tends to move simultaneously in some direction.

OR

The relationship between two variables is called Correlation.

e.g. As the number of internet users increase, the cinema viewers decrease, As the practical's marks increase, the marks of theory paper also increase etc.

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Positive Correlation OR Direct Correlation:

If both the variables move in same direction, the correlation is said to be Positive or Direct.

e.g. As the practical's marks increase, the marks of theory paper also increase etc.

Negative Correlation OR Inverse Correlation:

If both the variables move in opposite direction, the correlation is said to be Negative or Inverse.

e.g. As the number of internet users increase, the cinema viewers decrease etc.

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Note: *In Correlation, both the variables are treated as random.*

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Correlation and Causation Relationship:

A factor to consider is whether a causal relationship exists among two variables.

e.g. It is clear that a causal relationship exists in the example of steel output and labor input *i.e.* the number of workers will influence the number of tons of steel produced but *there is no causal relationship exit between use of apple and cancer rate.*

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PEARSON'S PRODUCT MOMENT CORRELATION COEFFICIENT OR COEFFICIENT OF CORRELATION:

A numerical measure of the strength of linear relationship is called as Coefficient of Correlation OR the Pearson's Product Moment Correlation Coefficient OR simply the Correlation Coefficient. It is denoted by r for sample data.

$$\text{i.e.} \quad r = \frac{\text{Cov.}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}} = \frac{S_{xy}}{\sqrt{S_x^2 \cdot S_y^2}}$$

$$\text{OR} \quad r = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2 \sum (Y - \bar{Y})^2}}$$

$$\text{OR} \quad r = \frac{\sum XY - \frac{\sum X \sum Y}{n}}{\sqrt{\left[\sum X^2 - \frac{(\sum X)^2}{n} \right] \left[\sum Y^2 - \frac{(\sum Y)^2}{n} \right]}}$$

$$\text{OR} \quad r = \frac{n \sum XY - \sum X \sum Y}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$$

$$\text{OR} \quad r = \frac{\sum XY - n\bar{X}\bar{Y}}{\sqrt{[\sum X^2 - n(\bar{X})^2][\sum Y^2 - n(\bar{Y})^2]}}$$

Note:

$$-1 \leq r \leq +1$$

- (i) If $r = +1$; there is Perfect Positive Correlation.
- (ii) If $r = -1$; there is Perfect Negative Correlation.
- (iii) If $r = 0$; there is no linear Correlation.

Example# 10: Compute the product moment coefficient of correlation between X and Y from the following data for the following data and interpret the result.

X	1	2	3	4	5
Y	2	5	3	8	7

Solution:

X	Y	XY	X^2	Y^2
1	2	2	1	4
2	5	10	4	25
3	3	9	9	9
4	8	32	16	64
5	7	35	25	49
$\sum X = 15$	$\sum Y = 25$	$\sum XY = 88$	$\sum X^2 = 55$	$\sum Y^2 = 151$

$$r = \frac{n \sum XY - \sum X \sum Y}{\sqrt{[n \sum X^2 - (\sum X)^2][n \sum Y^2 - (\sum Y)^2]}}$$

$$r = \frac{5(88) - (15)(25)}{\sqrt{[5(55) - (15)^2][5(151) - (25)^2]}}$$

$$r = \frac{440 - 375}{\sqrt{[275 - 225][755 - 625]}} = \frac{65}{\sqrt{(50)(130)}} = \frac{65}{\sqrt{6500}}$$

$$r = 0.81$$

Interpretation

e.g. $r = 0.81$ indicates that there is high positive Correlation between X and Y .

Example# 11: Find the correlation coefficient between X and Y given below;

X	5	12	4	16	18	21	22	23	25
Y	11	16	15	20	17	19	25	24	21

Example# 12: Calculate the correlation coefficient for the following data on supply (X) and demand (Y)

Supply (X)	400	200	700	100	500	300	600
Demand (Y)	50	60	20	70	40	30	10

Example# 13: Compute the correlation coefficient between X and Y represented in the following table;

X	2	4	5	6	8	11
Y	18	12	0	8	7	5

Example# 14: Find the product moment coefficient of correlation between traffic density and accident rate from the following information available. Find also the coefficient of Determination and interpret it.

Traffic Density	30	35	40	45	50	60	70	80	90
Accident Rate	2	4	5	5	8	15	24	30	32

Example# 15: Find the coefficient of correlation between persons employed and cloth manufactured in a textile mill. Interpret the result.

Persons employed	137	209	113	189	176	200	219
Cloth manufactured (000 yds)	23	47	22	40	39	51	49

D.Y.S.

PROPERTIES OF CORRELATION COEFFICIENT:

- (i) The Correlation Coefficient is symmetrical with respect to variables X and Y
i.e. $r_{xy} = r_{yx}$
- (ii) The Correlation Coefficient does not change by changing the origin or scale or both
i.e. $r_{uv} = r_{xy}$
Where $U = \frac{X - a'}{h}$ and $V = \frac{Y - a''}{k}$
- (iii) The Correlation Coefficient always lies between -1 and +1
i.e. $-1 \leq r \leq +1$
- (iv) The Correlation Coefficient is the Geometric mean of two regression coefficients
i.e. $r = \sqrt{b \times d}$
- (v) The Correlation Coefficient is a pure number i.e. it is independent from unit of measurement.

Example# 16: It is known that $r_{xy} = 0.70$ Find (i) r_{yx} (ii) r_{uv} where $u = 2x$ and $v = 3y$.

Solution: Given $r_{xy} = 0.7$

- (i) $r_{yx} = 0.70$

Because correlation coefficient is symmetrical with respect to variables x and y i.e.

$$r_{xy} = r_{yx}$$

(ii) $r_{uv} = 0.70$

Because correlation coefficient does not change by changing the origin and scale i.e.

$$r_{uv} = r_{xy}$$

Note:

- (i) Both the regression coefficients always have the same signs for the same data.
 (ii) Correlation Coefficient has the same sign as the regression coefficients.

Example# 17: If two regression coefficients have the values (i) 0.75 and 0.23 (ii) -0.80 and -0.07, then find r ?

Solution: Given $r_{xy} = 0.7$

(i) Given $b = 0.75$ and $d = 0.23$

As we know that

$$r = \sqrt{b \times d}$$

$$r = \sqrt{0.75 \times 0.23} = 0.415$$

(ii) Given $b = -0.80$ and $d = -0.07$

As we know that

$$r = \sqrt{b \times d}$$

$$r = \sqrt{-0.80 \times -0.07} = -0.24$$

RANK CORRELATION:

The relationship between the ranks of two individuals is called Rank Correlation.

SPEARMAN'S RANK CORRELATION COEFFICIENT:

A numerical measure of the strength of linear relationship between the ranks of two individuals is called Rank Correlation Coefficient **OR** Spearman's Rank Correlation Coefficient. It is denoted by r_s for sample data.

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

Where $d = X - Y$ **OR** $d = Y - X$

and "d" stand for difference of Ranks.

Example# 18: Find the coefficient of rank correlation from the following two sets of values.

A	7.4	9.0	11.0	2.5	4.6	6.5
B	8.5	6.1	2.4	6.7	12.6	3.3

Solution:

		Ranks		$d = X - Y$	d^2
A	B	X	Y		
7.4	8.5	3	2	1	1
9.0	6.1	2	4	-2	4
11.0	2.4	1	6	-5	25
2.5	6.7	6	3	3	9
4.6	12.6	5	1	4	16
6.5	3.3	4	5	-1	1
					$\Sigma d^2 = 56$

$$r_s = 1 - \frac{6 \Sigma d^2}{n(n^2 - 1)}$$

$$r_s = 1 - \frac{6(56)}{6(6^2 - 1)}$$

$$r_s = 1 - \frac{336}{210}$$

$$r_s = 1 - 1.6$$

$$r_s = -0.6$$

Example# 19: The following table shows how 10 students, arranged in alphabetical order, were ranked according to their achievements in both laboratory and lecture portions of a Statistics course. Find the coefficient of rank correlation.

Laboratory	8	3	9	2	7	10	4	6	1	5
Lecture	9	5	10	1	8	7	3	4	2	6

Solution:

Laboratory X	Lecture Y	$d = X - Y$	d^2
8	9	-1	1
3	5	-2	4
9	10	-1	1
2	1	1	1
7	8	-1	1
10	7	3	9
4	3	1	1
6	4	2	4
1	2	-1	1
5	6	-1	1
			$\Sigma d^2 = 24$

$$r_s = 1 - \frac{6 \Sigma d^2}{n(n^2 - 1)}$$

$$r_s = 1 - \frac{6(24)}{10(10^2 - 1)} = 1 - \frac{144}{990}$$

$$r_s = 1 - 0.145$$

$$r_s = 0.855$$

Example# 20: Find the coefficient of rank correlation from the following ranking of 10 students in Statistics and Mathematics.

Statistics(x)	1	2	3	4	5	6	7	8	9	10
Mathematics (y)	2	4	3	1	7	5	8	10	6	9

D.Y.S.

Example# 21: Ten competitors in a beauty contest are ranked by three judges in the following order.

First Judge	1	6	5	10	3	2	4	9	7	8
Second Judge	3	5	8	4	7	10	2	1	6	9
Third Judge	6	4	9	8	1	2	3	10	5	7

Use the rank correlation coefficient to discuss which pair of judges has the nearest approach to common tastes in beauty.

Solution:

Let 1st Judge = X

Let 2nd Judge = Y

Let 3rd Judge = Z

X	Y	Z	$d_{XY} = X - Y$	d_{XY}^2	$d_{XZ} = X - Z$	d_{XZ}^2	$d_{YZ} = Y - Z$	d_{YZ}^2
1	3	6	-2	4	-5	25	-3	9
6	5	4	1	1	2	4	1	1
5	8	9	-3	9	-4	16	-1	1
10	4	8	6	36	2	4	-4	16
3	7	1	-4	16	2	4	6	36
2	10	2	-8	64	0	0	8	64
4	2	3	2	4	1	1	-1	1
9	1	10	8	64	-1	1	-9	81
7	6	5	1	1	2	4	1	1
8	9	7	-1	1	1	1	2	4
				$\sum d_{XY}^2 = 200$		$\sum d_{XZ}^2 = 60$		$\sum d_{YZ}^2 = 214$

$$r_{s(XY)} = 1 - \frac{6 \sum d_{XY}^2}{n(n^2 - 1)} = 1 - \frac{6(200)}{10(10^2 - 1)} = -0.21$$

$$r_{s(XZ)} = 1 - \frac{6 \sum d_{XZ}^2}{n(n^2 - 1)} = 1 - \frac{6(60)}{10(10^2 - 1)} = 0.64$$

$$r_{s(YZ)} = 1 - \frac{6 \sum d_{YZ}^2}{n(n^2 - 1)} = 1 - \frac{6(214)}{10(10^2 - 1)} = -0.30$$

Therefore 1st and 3rd judge have nearest approach in common taste, because they have highest correlation between them.

Example# 22: In a painting competition, various entries are ranked by three judges. Use Spearman's rank correlation coefficient to discuss which pair of judges has the nearest approach to common taste.

Judge X	5	2	6	8	1	7	4	9	3	10
Judge Y	1	7	6	10	4	5	3	8	2	9
Judge Z	6	4	9	8	1	2	3	10	5	7

D.Y.S.

.....

Rank Correlation Coefficient For Tied Values:

$$r_s = 1 - \frac{6 \text{Adj.} \sum d^2}{n(n^2 - 1)}$$

Where $\text{Adj.} \sum d^2 = \sum d^2 + T$

and $T = \frac{1}{12}(t^3 - t)$ where t = number of tied values.

.....

Example# 23: Find the coefficient of rank correlation from the following two sets of values.

X	50	43	50	40	60	40	40	55
Y	90	95	112	120	95	170	100	120

Solution:

X	Y	Ranks		d = a - b	d²
		a	b		
50	90	$\frac{3+4}{2} = 3.5$	8	-4.5	20.25
43	95	5	$\frac{6+7}{2} = 6.5$	-1.5	2.25
50	112	3.5	4	-0.5	0.25
40	120	$\frac{6+7+8}{3} = 7$	$\frac{2+3}{2} = 2.5$	4.5	20.25
60	95	1	6.5	-5.5	30.25
40	170	7	1	6	36
40	100	7	5	2	4
50	90	2	2.5	-0.5	0.25
					$\sum d^2 = 113.5$

$$r_s = 1 - \frac{6 \text{Adj.} \sum d^2}{n(n^2 - 1)}$$

$$T_1 = \frac{1}{12}(t_1^3 - t) = \frac{1}{12}(2^3 - 2) = 0.5$$

$$T_2 = \frac{1}{12}(t_2^3 - t) = \frac{1}{12}(3^3 - 3) = 20$$

$$T_3 = \frac{1}{12}(t_3^3 - t) = \frac{1}{12}(2^3 - 2) = 0.5$$

$$T_4 = \frac{1}{12}(t_4^3 - t) = \frac{1}{12}(2^3 - 2) = 0.5$$

Now $T = T_1 + T_2 + T_3 + T_4$

$$T = 0.5 + 2.0 + 0.5 + 0.5 = 3.5$$

$$Adj. \sum d^2 = \sum d^2 + T = 113.5 + 3.5 = 117.0$$

$$r_s = 1 - \frac{6Adj. \sum d^2}{n(n^2 - 1)}$$

$$r_s = 1 - \frac{6(117.0)}{8(8^2 - 1)} = 1 - \frac{702}{504}$$

$$r_s = -0.39$$

.....

Example# 24: Find the coefficient of rank correlation from the following ranks.

X	8	3	6.5	3	6.5	9	3	1	5
Y	8	9	6.5	2.5	4	5	6.5	1	2.5

D.Y.S.

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